ArtFusion: An Arbitrary-Reference Spatiotemporal Fusion Model for Seamless Remote Sensing Image Reconstruction

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***Abstract*—Spatiotemporal fusion (STF) represents a vital solution for continuous high-resolution Earth observation. Despite the increasing abundance of historical satellite archives, most existing STF methods are constrained by a limited number of auxiliary reference images (typically one or two), leading to underutilized multitemporal information and reduced reconstruction reliability, particularly when References are compromised by cloud contamination or abrupt land-surface variations. To address these limitations, this study proposes an arbitrary-reference STF (ArtFusion) model. The architecture incorporates an efficient contrast-aware hybrid block (ECHB) for deep feature extraction, an explicit temporal information encoder (ETIE) to utilize acquisition metadata, and a multihead cross-reference attention fusion (MCAF) module designed to facilitate the integration of an arbitrary number of reference images. A comprehensive evaluation across 24 experimental cases in three representative study regions demonstrates that ArtFusion consistently outperforms four state-of-the-art (SOTA) benchmarks [multilevel feature fusion with generative adversarial network (MLFF-GAN), C-ROBOT, RealFusion, and frequency-selected differential fusion transformer (FSDFormer)]. Compared with the best-performing baseline among the four competing methods, ArtFusion achieves average improvements of 6.8% in spectral accuracy [root-mean-square error (RMSE)] and 15.3% in spatial accuracy [improved edge difference metric (iEDGE)]. Notably, high reliability is maintained across four challenging scenarios: rapid phenological changes, drastic morphological variations, highly heterogeneous landscapes, and frequent cloud contamination, while also demonstrating strong cross-regional transferability. Despite its superior performance, ArtFusion has an extremely compact structure with 0.19 million trainable parameters, only 2.2% of MLFF-GAN's parameters. This work demonstrates the potential of leveraging multiple reference images to push the boundaries of STF accuracy, rather than merely increasing model complexity. This flexible multireference fusion scheme provides a promising pathway for robust, large-scale Earth observation in cloudy and dynamically changing landscapes. The source code is available at: https://github.com/Andy-cumt/ArtFusion-STF**

***Index Terms*—Arbitrary-reference fusion, attention mechanism, deep learning, spatiotemporal fusion (STF).**

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I. Introduction

SPATIOTEMPORAL fusion (STF) is recognized as a promising technical pathway for reconstructing dense time-series images with high spatial resolution. This approach has been extensively applied in fields with high time-sensitivity requirements, such as agricultural monitoring [1], [2], and disaster assessment [3], [4]. Advancements in this domain are pivotal for enhancing the continuity and precision of global environmental monitoring. [5], [6], [7]. Typically, STF integrates two types of imagery: the first category features a narrow swath and a long revisit cycle but fine spatial resolution (e.g., Sentinel-2, Landsat series, hereafter referred to as fine images), while the second offers a wide swath and a short revisit cycle but coarse spatial resolution (e.g., Sentinel-3, MODIS, hereafter referred to as coarse images). By fusing these multisource images, STF reconstructs a synthetic sequence that is both temporally dense and spatially fine.

Over the past decade, STF technology has garnered considerable attention within the remote sensing community. Based on their underlying principles, existing methods can be broadly categorized into traditional methods and deep learning-based methods. Traditional methods can be further subdivided into filtering-based, unmixing-based, traditional machine learning, and hybrid methods.

Among these, filtering-based methods generate fine imagery by applying weighted filters that account for the spatiotemporal correlations between reference and prediction phases. One of the most seminal methods is the spatial and temporal adaptive reflectance fusion model (STARFM), proposed by Gao et al. [8]. This framework has since inspired the development of numerous variants. For instance, the enhanced STARFM (ESTARFM) proposed by Zhu et al. [9], which

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improved fusion performance in heterogeneous regions by introducing a linear correlation coefficient to refine weight assignment. Liu et al. [10] further enhanced the robustness of ESTARFM by adaptively determining moving window sizes through the consideration of spatial heterogeneity. In addition, Wang and Atkinson [11] employ local linear regression, spatial filtering based on spectrally similar neighbor pixels, and residual distribution, demonstrating superior information recovery in scenarios with abrupt phenological changes compared to STARFM. Despite these incremental refinements, filtering-based methods primarily rely on local pixels and operate on a stationarity assumption, which assumes that the relationship between neighboring pixels remains invariant. This localized perspective hinders their ability to reconstruct abrupt land-surface variations.

Unmixing-based methods are built upon the principle of the linear spectral mixture model, assuming that pixels of the same class share similar spectral characteristics or exhibit identical temporal variations [12]. These methods achieve super-resolution by solving linear mixing equations using global or local pixel information. The multisensor multiresolution technique (MMT) proposed by Zhukov et al. [13] was the first unmixing-based STF method, and subsequent methods have extensively built upon and refined this framework. For example, the object-based spatial and temporal vegetation index unmixing model (OB-STVIUM) proposed by Lu et al. [14] utilized multiscale segmentation to better define endmember fractions, thereby improving prediction accuracy. Wang et al. [15] introduced new constraints to the unmixing objective function to mitigate blocking effects. Peng et al. [16] enhanced unmixing accuracy by incorporating geospatial weighting and soft classification. While these methods excel at preserving heterogeneous information, they often lack intraclass variability, meaning their accuracy degrades rapidly if land-cover types change between the two observation phases.

Machine learning methods involve modeling via classical statistical learning techniques such as random forest and dictionary pair learning. For instance, the sparse representation-based spatiotemporal reflectance fusion model (SPSTFM), proposed by Huang and Song [17], first introduced dictionary pair learning to STF. The ROBust OpTimization-based (ROBOT) fusion model, proposed by Chen et al. [18], utilizes the least absolute shrinkage and selection operator (Lasso) for long-term time-series modeling, showing significant advantages over existing methods that rely on single-pair reference images. However, with the advent of deep learning, traditional machine learning struggles to compete in feature extraction, leading to its gradual replacement in research prominence.

Hybrid methods aim to leverage the complementary strengths of multiple techniques. A representative example is the flexible spatiotemporal data fusion (FSDAF) method proposed by Zhu et al. [19], which integrates unmixing, filtering, and spatial interpolation techniques to simultaneously recover phenological dynamics and land-cover changes. Building on this, Li et al. [20] enhanced FSDAF by incorporating subpixel mapping technology, enabling improved representation of fine-scale features. Similarly, Guo et al. [21] proposed an object-based spatial unmixing model (OBSUM), which integrates the segment anything model (SAM) for large-scale image segmentation with spectral unmixing to refine agricultural parcel-level STF accuracy. In addition, Zhou and Zhong [22] proposed a Kalman filter reflectance fusion model (KFRFM) that couples Kalman filtering with linear unmixing to assess uncertainty in fusion results. Despite their integrative advantages, hybrid methods remain constrained by manual feature engineering challenges, limited capacity for capturing deep-level representations, and computational inefficiency arising from complex model nesting.

With the rapid advancement of deep learning in computer vision, numerous network models specifically designed for STF tasks have emerged and gained widespread attention [23], [24]. The first deep learning-based model was introduced by Song et al. [25], followed by Tan et al. [26] in the same year. While early models were limited to simple stacks of convolutional neural networks (CNNs), they demonstrated greater potential than traditional methods. As deep learning evolved, new architectures were rapidly adapted to STF, leading to a surge in innovative methods and significant performance gains. Examples include the introduction of generative adversarial networks (GANs) by Zhang et al. [27], Tan et al. [28], [29], the application of transformers by [30], the integration of wavelet transforms by [31], and more recently, the incorporation of diffusion models [32], task decoupling strategies [33] and multimodal fusion [34]. By leveraging hierarchical deep architectures to distill high-dimensional features, these methods can fit intricate surface changes and exhibit higher robustness to cloud contamination and noise [35]. Despite their superior accuracy and potential, current deep learning-based STF research still faces two major shortcomings as follows.

1) Insufficient utilization of multitemporal historical data leads to limited accuracy and robustness. Periodic variations in surface information are ubiquitous, such as phenological cycles and seasonal fluctuations in wetland water bodies. Furthermore, historical imagery archives capture abrupt events, including landslides and floods, indicating a strong spatiotemporal correlation across different observation phases. However, the majority of current deep learning-based STF methods rely on only one or two fine reference images to accommodate scenarios with scarce auxiliary data, primarily focusing on increasing architectural complexity to enhance precision. This paradigm is increasingly incompatible with the current abundance of historical satellite data. The reliance on limited auxiliary information, compounded by the substantial scale gap between heterogeneous sensors, presents a significant technical challenge. For instance, the resolution difference between Landsat and MODIS often exceeds 16×, which far surpasses the 2× to 8× magnification typically addressed in computer vision super-resolution tasks [36], [37], [38], [39]. Because STF must simultaneously account for extreme scale differences, cloud contamination, and sensor noise, it is effectively a highly ill-posed inverse problem that combines super-resolution, de-clouding, and denoising. These factors collectively create a performance ceiling that hinders existing STF methods from achieving

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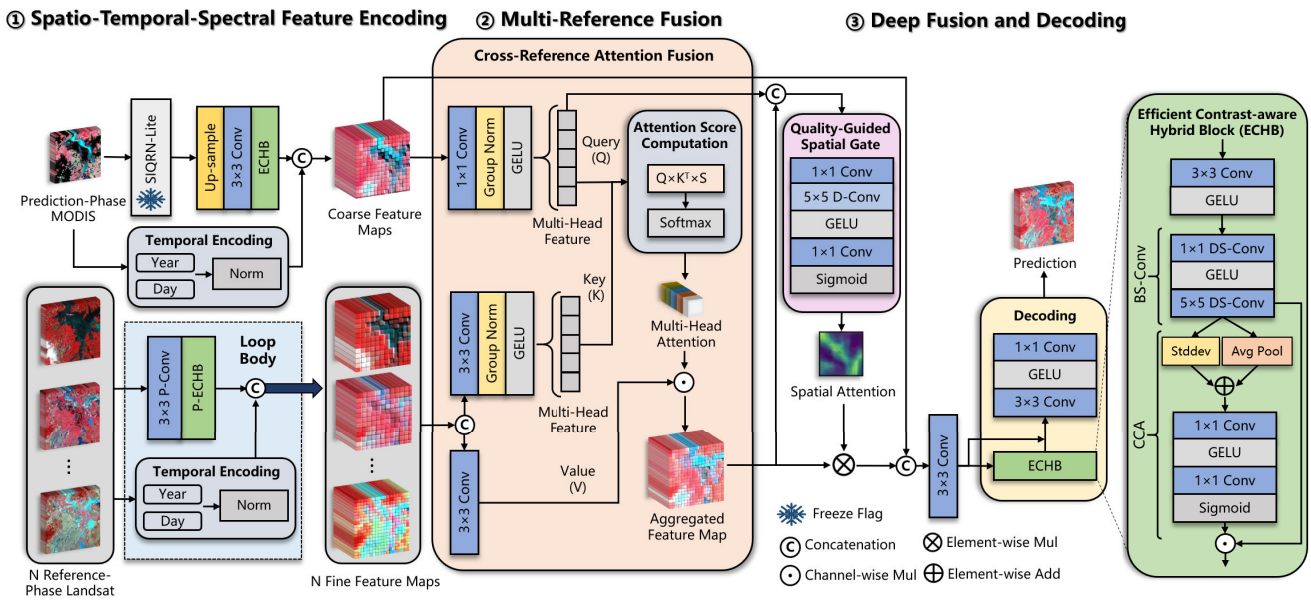


Fig. 1. Architecture of the proposed ArtFusion model.

further breakthroughs in reconstruction accuracy. Fortunately, the inherent correlation between reference and target images in remote sensing suggests that incorporating additional multitemporal data can theoretically mitigate this ill-posedness [40], [41]. By leveraging images across different seasons, the model can better capture land-cover dynamics and seasonal trajectories [42]. This perspective has been validated by multireference image super-resolution tasks in computer vision [43], as well as recent traditional STF methods, such as ROBOT [18] and its variants [43], which effectively improve accuracy by modeling long-term sequences via Lasso regression. Consequently, the efficient and reliable integration of rich historical data is essential to maximize the potential of STF and achieve higher reconstruction reliability.

2) Neglect of explicit temporal information hinders the efficient fusion of highly correlated information. Existing STF paradigms primarily rely on image-based feature extraction, largely neglecting the explicit metadata regarding image acquisition dates. Temporal information encapsulates patterns of surface change (e.g., seasonal phenology) and reflects the temporal intervals between reference and target images, which are critical for estimating the likelihood of surface change. Theoretically, the introduction of temporal features can effectively refine fusion accuracy. To address the aforementioned limitations, this study proposes an arbitrary-reference STF (ArtFusion) model. By integrating a lightweight feature extraction framework with explicit timestamp encoding and cross-reference attention mechanisms, the model enables the fusion of an arbitrary number of reference images for seamless reconstruction. This design facilitates the comprehensive exploitation of extensive historical archives, ensuring robust and reliable reconstruction even under complex environmental conditions. Notably, ArtFusion achieves superior accuracy with a highly compact architecture of only 0.19 million parameters, significantly reducing the computational overhead compared to existing networks and eliminating the requirement for manual reference selection. To the best of the available knowledge, this represents the first STF network to support a flexible number of auxiliary inputs and incorporate explicit temporal metadata. The remainder of this article is organized as follows. Section II details the methodology of the ArtFusion model. Section III describes the datasets and experimental design utilized for validation. Section IV presents the experimental results and comprehensive analysis, including standard evaluations and ablation studies across various heterogeneous study areas. Section V provides an in-depth discussion regarding the impact of input image quantity, computational efficiency, and a comparison with representative STF frameworks, alongside the limitations of the proposed work. Finally, Section VI summarizes the findings and concludes this article.

II. Methodology

The architecture of the proposed ArtFusion is illustrated in Fig. 1. The framework comprises three integral components: 1) spatiotemporal–spectral feature encoding; 2) multireference fusion; and 3) deep fusion and decoding reconstruction. In this study, Landsat-8 and MODIS reflectance products serve as the fine and coarse imagery, respectively, selected for their extensive historical archives and the strong ill-posedness arising from their substantial scale differences, which provides a rigorous test of method reliability. ArtFusion is also applicable to other satellite data combinations, such as Landsat-8/9 with Sentinel-2 and Sentinel-2 with Sentinel-3.

A. Spatiotemporal–Spectral Feature Encoding

Reliable feature extraction from multisource cross-spatiotemporal imagery is the foundation of STF. As

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illustrated in Fig. 1, ArtFusion employs a dual-stream encoding architecture with quasi-symmetrical fine (Landsat) and coarse (MODIS) branches. Specifically, the MODIS stream includes a dedicated restoration step. Because the MODIS image at the prediction phase provides indispensable spectral and spatial guidance, its input quality directly dictates the reliability of the fusion output. To ensure high-fidelity guidance, a frozen, pretrained lightweight variant of the spatiotemporal information injection-based quality restoration network (SIQRN-lite) [33] is integrated to refine the MODIS product (refer to the Appendix for detailed specifications). This module operates as an independent, nontrainable unit to mitigate spectral discrepancies and align MODIS observations more closely with Landsat sensor characteristics. Subsequently, the restored image is upsampled to the Landsat spatial size and projected into a high-dimensional feature space via convolution.

To address the substantial computational overhead of multitemporal auxiliary sequences, this study proposes a novel, efficient contrast-aware hybrid block (ECHB). As shown in Fig. 1 (right panel), ECHB balances representational capacity and computational efficiency by integrating blueprint separable convolution (BS-Conv) [45] to mitigate parameter redundancy. Unlike standard depthwise separable convolutions, BS-Conv prioritizes interchannel over intrakernel spatial correlations, enabling significant receptive field expansion via large (5×5) kernels with minimal parameter overhead. To further enhance feature distinctiveness, ECHB incorporates the contrast-aware channel attention (CCA) mechanism [46], which captures high-frequency details neglected by standard pooling. Given an input feature tensor $X \in \mathbb{R}^{C\times H\times W}$, the channel-wise statistics $z \in \mathbb{R}^C$ are aggregated using both mean and standard deviation to characterize texture contrast

$$z_c = \text{AvgPool}(X_c) + \sqrt{\text{Var}(X_c) + \epsilon} \tag{1}$$

where ϵ is a small constant for numerical stability. Subsequently, an attention map $M \in \mathbb{R}^{C\times 1\times 1}$ is generated through a bottleneck structure consisting of dimension reduction, activation, and restoration layers. These learned weights are then utilized to adaptively recalibrate the input features via element-wise multiplication. This contrast-aware design enables the network to effectively emphasize high-frequency components, thereby recovering fine-grained textures and boundary details critical for the STF task.

Beyond visual features, this study introduces an explicit temporal information encoder (ETIE) to handle time-series dependencies. The acquisition dates are parsed from image metadata to calculate the normalized year and the day of year (DOY). To preserve the cyclical nature of seasonal changes, the DOY is encoded using sinusoidal functions. For each image at time t, a 3-D temporal vector v_t is generated

$$v_t = \left[\hat{y}, \sin\left(\frac{2\pi d}{365}\right), \cos\left(\frac{2\pi d}{365}\right)\right] \tag{2}$$

where $\hat{y}$ is the normalized year and d is the DOY. This vector is spatially expanded and concatenated with the extracted image features, guiding the network to prioritize reference images that are temporally correlated with the prediction phase.

Crucially, to maintain architectural compactness when processing multitemporal Landsat sequences, a parameter-sharing strategy is employed. The convolutional layers, ECHB, and ETIE within the Landsat stream share identical weights (denoted as P-Conv and P-ECHB) and are applied recurrently to extract features from each reference phase.

B. Multireference Fusion

To accurately capture beneficial features from multiple reference images, the multihead cross-reference attention fusion (MCAF) module is proposed. Drawing inspiration from the transformer architecture [47], MCAF employs multihead attention to partition feature channels into distinct subspaces, enabling parallel pixel-wise attention computation across these subspaces. This design facilitates the simultaneous modeling of diverse spatiotemporal dependencies, thereby achieving superior feature aggregation compared to single-head counterparts. Denoting the features from the prediction-phase MODIS as C_{pre} and the set of reference-phase Landsat features as $F_{ref} = \{F^1_{ref}, \ldots, F^N_{ref}\}$. These are projected into Query (Q), Key (K), and Value (V) embeddings

$$Q = \phi_q(C_{pre}), \quad K = \phi_k(F_{ref}), \; V = \phi_v(F_{ref}) \tag{3}$$

where $\phi(\cdot)$ represents linear projections implemented via convolutions. Since the Q, K, and V projections are independently applied to each reference image and the subsequent attention computation is performed along the temporal dimension, the proposed formulation is inherently independent of the number of reference images. Consequently, the network is not restricted to a fixed input size during inference and can naturally accommodate arbitrary numbers of reference images, including configurations not explicitly encountered during training.

The attention mechanism utilizes the prediction-phase MODIS features as a prior to adaptively weight the reference features. To maximize computational efficiency, the pixel-wise dot-product similarity is calculated directly using the Einstein summation convention. The fused feature map F_{fused} is computed as

$$F_{fused} = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{4}$$

where d_k is the scaling factor. Crucially, the Softmax function is applied along the temporal dimension, allowing the model to dynamically aggregate the most relevant feature information from the multiple reference images for each pixel location.

C. Deep Fusion and Decoding Reconstruction

Following the initial fusion of multireference features under the guidance of the prediction-phase MODIS information, a deeper synthesis is required to reconcile potential discrepancies between multisource representations. To this end, a quality-guided spatial gate (QGSG) was developed to further refine the composite features. This module functions as a spatial attention mechanism designed to handle regions with high uncertainty, such as those affected by cloud contamination or rapid land-cover changes.

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The QGSG generates a spatial gating map G by processing the concatenated features of the prediction-phase and the fused references. The operation incorporates a large-kernel depth-wise convolution (D−Conv) to capture contextual information

$$G = \sigma\left(\mathrm{Conv}_{1\times1}\left(\delta\left(\mathrm{D}-\mathrm{Conv}_{5\times5}\left(\mathrm{Conv}_{1\times1}([C_{pre}, F_{fused}])\right)\right)\right)\right) \tag{5}$$

where $[\cdot,\cdot]$ denotes concatenation. The fused features are then refined by this gate, suppressing unreliable information while amplifying high-quality regions. σ is the Sigmoid activation ensuring values in $(0, 1)$, and δ is the GELU activation. The final reconstruction is obtained by decoding the combined features through a decoder composed of convolutional layers and ECHB

$$P_{Art} = \mathcal{D}\left(\left[C_{pre}, F_{fused} \otimes G\right]\right) \tag{6}$$

where $\mathcal{D}(\cdot)$ represents the decoding network and P_{Art} is the final prediction.

D. Loss Function

ArtFusion is optimized through a hybrid loss formulation that jointly supervises spectral fidelity, spatial structure, and perceptual similarity [3]. Prior to loss computation, cloud masks derived from the prediction-phase Landsat ground truth F_P are applied to both the fused output and the ground truth itself.

The spectral component L_{spec} employs the Charbonnier loss [8], providing a differentiable smooth approximation of the L1 norm. This term operates per spectral band b with adaptive weighting factors $\omega^{(b)}$ to balance contributions across wavelengths

$$\mathcal{L}_{\text{spec}} = \frac{1}{B}\sum_{b=1}^{B}\sqrt{\left[\left|P_{\text{Art}}^{(b)} - F_P^{(b)}\right|\cdot\omega^{(b)}\right]^2 + \varepsilon^2} \tag{7}$$

where $\omega^{(b)}$ represents band-specific scaling coefficients derived from per-band mean statistics, B denotes the total number of spectral channels, and $\varepsilon = 10^{-5}$ ensures numerical stability.

The spatial regularization $\mathcal{L}_{\text{edge}}$ enforces edge consistency by quantifying the normalized gradient magnitude discrepancy between prediction and ground truth. This employs the Roberts operator, a diagonal-gradient detection kernel that approximates 2-D spatial derivatives through discrete directional differences

$$\boldsymbol{L}_{\text{edge}} = \mathbb{E}\left[\frac{|\mathrm{Edge}(P_{\text{Art}}) - \mathrm{Edge}(F_P)|}{\mathrm{Edge}(P_{\text{Art}}) + \mathrm{Edge}(F_P) + \varepsilon}\right] \tag{8}$$

where $\mathrm{Edge}(\cdot)$ denotes Roberts edge extraction and $\mathbb{E}[\cdot]$ computes the pixel-wise average.

The perception loss $\mathcal{L}_{\text{perc}}$ is defined as the complement of multiscale structural similarity (MS-SSIM) [9], which assesses structural correspondence across multiple resolutions to align with human visual perception

$$\mathcal{L}_{\text{perc}} = 1 - \mathrm{MS\text{-}SSIM}(P_{\text{Art}}, F_P). \tag{9}$$

The weighting coefficients were determined according to two principles: 1) maintaining the three loss terms at comparable magnitudes to avoid optimization imbalance and 2) assigning greater importance to $\mathcal{L}_{\text{perc}}$ and $\mathcal{L}_{\text{spec}}$, which better reflect overall reconstruction quality and spectral fidelity, than to $\mathcal{L}_{\text{edge}}$, which primarily emphasizes boundary preservation. Based on these considerations, λ_{spec}, λ_{edge}, and λ_{perc} were empirically set to 30, 0.5, and 9, respectively. This design ensures that spectral fidelity remains the primary optimization objective for quantitative remote sensing applications. The resulting composite loss function is defined as

$$\mathcal{L}_{\text{Art}} = \lambda_{\text{spec}}\mathcal{L}_{\text{spec}} + \lambda_{\text{edge}}\mathcal{L}_{\text{edge}} + \lambda_{\text{perc}}\mathcal{L}_{\text{perc}}. \tag{10}$$

III. Study Areas and Experimental Design

A. Study Areas

To rigorously benchmark the proposed method, experiments were conducted across two geographically and temporally challenging regions characterized by distinct land cover dynamics.

The first region, depicted in Fig. 2(a), encompasses the Poyang Lake Wetland in northern Jiangxi, China (115.6803°E, 28.4214°N). This site integrates a complex mosaic of Poyang Lake's wetland ecosystem, peripheral aquaculture ponds, fragmented agricultural parcels, vegetated foothills, and the heterogeneous urban fabric of Nanchang. Seasonal hydrological cycles drive substantial intra-annual lake extent fluctuations. Persistent cloud contamination, attributable to the subtropical monsoonal climate and high evapotranspiration from the lake surface, frequently compromises optical observations. Time-series imagery spanning 2014–2021 was acquired from Landsat-8 and MODIS sensors, yielding 48 image pairs (32 for training, 8 for validation, 8 for testing).

The second region, illustrated in Fig. 2(b), covers the Western Kansas Agricultural Zone straddling western Kansas and eastern Colorado, USA (102.4169°W, 38.6822°N). Situated within the Central Great Plains, this area is dominated by extensively managed circular and rectangular center-pivot irrigated fields exhibiting pronounced seasonal phenological cycles and land-use transitions driven by crop rotation and harvest schedules. Landsat-8 OLI surface reflectance products (2017–2021) with cloud cover below 20% served as fine references, paired with contemporaneous MODIS acquisitions as coarse inputs. A total of 61 image pairs were compiled, allocated as 41 training, 10 validation, and 9 testing samples.

The third region, shown in Fig. 2(c), is located on the southern flank of the Taihang Mountains at the junction of Shanxi, Henan, and Hebei Provinces, China (113.4933°E, 35.2290°N). The area comprises mountainous terrain, extensive croplands, and urban settlements. In July 2021, an extreme rainfall event triggered widespread flooding, leaving large agricultural areas submerged for several months. A total of 42 image pairs were collected, including 28 training, 7 validation, and 7 testing samples. Notably, the training set contains no flood-related observations, providing a challenging benchmark for assessing a model's ability to predict previously unseen, large-scale, and highly nonstationary surface changes.

All datasets were accessed and preprocessed via Google Earth Engine, including co-registration, extent clipping, and dimension standardization to 3200 × 3200 pixels. Both the Landsat-8 reflectance product and the MCD43A4 product

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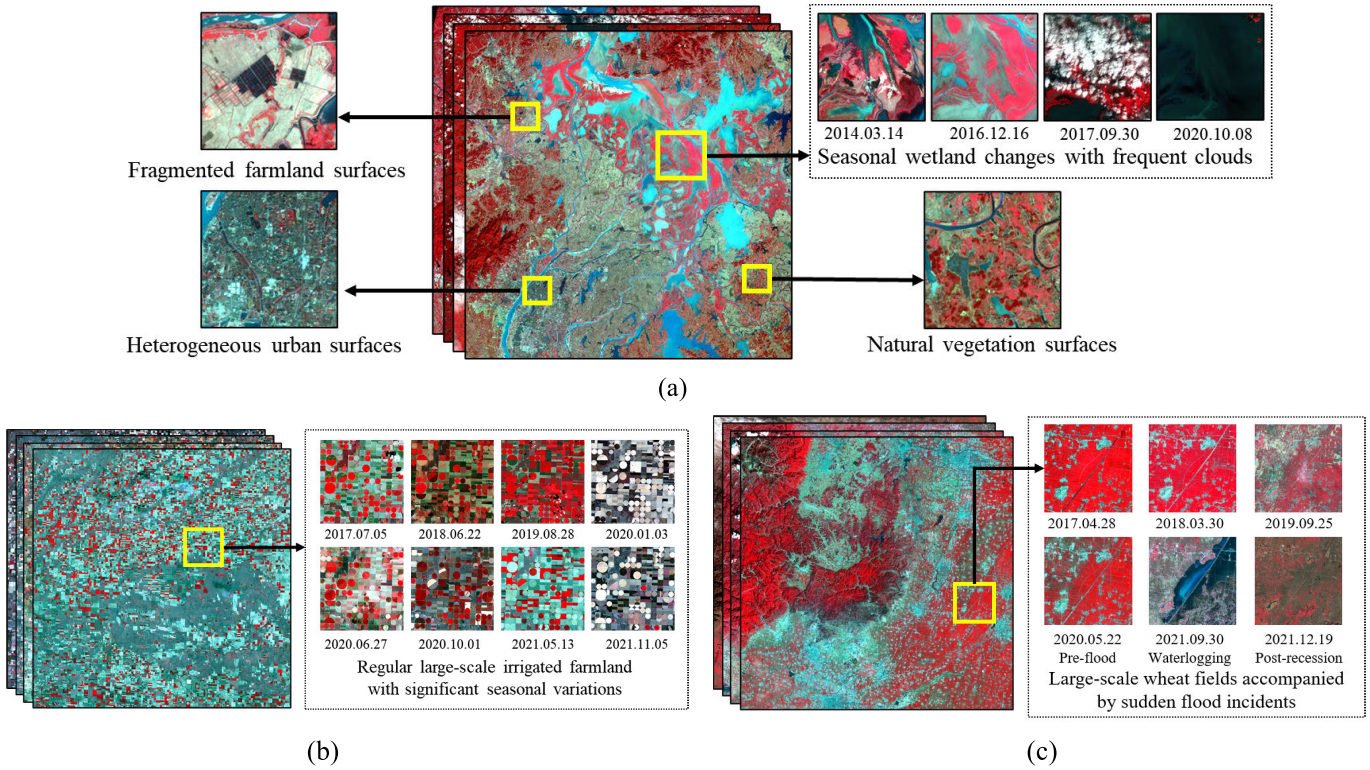


Fig. 2. Typical Landsat images of three study areas. (a) Poyang Lake Wetland. (b) Western Kansas Agricultural Zone. (c) Southern Taihang Mountains Area.

stacks comprise six bands: Blue, Green, Red, NIR, SWIR1, and SWIR2.

B. Experimental Design

To comprehensively evaluate the performance of the proposed method, several state-of-the-art (SOTA) STF methods were selected as benchmarks. Specifically, the first benchmark is RealFusion [33], a representative deep learning-based SOTA STF method. Based on a task-decoupling strategy, RealFusion employs the full SIQRN model for MODIS reconstruction and incorporates annual median-synthesized fine-resolution NDVI to mitigate cloud-induced information loss and enhance phenological prediction. The second benchmark is ROBOT [18], a traditional SOTA STF method capable of integrating an arbitrary number of MODIS-Landsat pairs. To ensure a fair comparison regarding long-term sequence processing capabilities, the MODIS products input to ROBOT were also corrected using SIQRN-lite, hereafter referred to as C-ROBOT.

In addition, two representative single-reference-pair fusion methods were included as controls. The first is frequency-selected differential fusion transformer (FSDFormer), a cutting-edge transformer-based SOTA proposed in 2025 [50] that achieves linear computational complexity while enhancing high-frequency detail reconstruction and explicit spatiotemporal dependency modeling. The second is the multilevel feature fusion with GAN (MLFF-GAN) [29], which has demonstrated competitive performance across various STF scenarios [31]. The input configurations for each method are summarized in Table I.

Except for FSDFormer, which was evaluated using an input size of 256 × 256 due to architectural constraints, all models were trained and tested with 400 × 400 inputs. For ArtFusion, the initial learning rate was set to 110^{-3}, and the Adam optimizer was employed for gradient descent-based parameter updates. During the training stage, nine reference images were randomly sampled for each epoch to enhance the model's generalization across various temporal distances. In the inference stage, 15, 11, and 11 reference images were employed for the Poyang Lake Wetland, the Western Kansas Agricultural Zone, and the Southern Taihang Mountains Area, respectively. To ensure a fair comparison, four deep learning-based methods followed an identical training strategy where areas contaminated by clouds in the prediction phase were excluded from the loss calculation, and the coarse image of

TABLE I
INPUT CONFIGURATIONS FOR DIFFERENT STF METHODS

Method	Fine Imagery	Coarse Imagery	
	Reference-Phase(s)	Reference-Phase(s)	Prediction Phase
MLFF-GAN	Landsat Reflectance Product (single-temporal)	MCD43A4 (single-temporal)	MCD43A4
C-ROBOT	Landsat Reflectance Product (long-term temporal)	MCD43A4 (long-term temporal)	MCD43A4
RealFusion	Landsat NDVI (annual median composite); Landsat Reflectance Product (single-temporal)	MCD43A4 (single-temporal)	MCD43A4
FSDFormer	Landsat Reflectance Product (single-temporal)	MCD43A4 (single-temporal)	MCD43A4
ArtFusion	Landsat Reflectance Product (long-term temporal)	None	MCD43A4

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the prediction phase was randomly selected for training. All experiments were performed on a workstation equipped with an NVIDIA GeForce RTX 5070 Ti GPU and an Intel Ultra 7265 K CPU.

The experimental framework consists of routine experiments and ablation studies. Routine experiments assess accuracy through qualitative visualization and quantitative metrics. Given the substantial data volume, including 17 experimental cases and 3200 × 3200-pixel full-scene tiles, qualitative analysis focuses on local enlargements within four representative scenarios for clear, detailed comparison: 1) areas with rapid phenological changes; 2) drastic morphological variations; 3) highly heterogeneous landscapes; and 4) frequent cloud contamination. Quantitative evaluation follows the all-round performance assessment (APA) framework [51]. Spectral accuracy is assessed using root-mean-square error (RMSE), while spatial accuracy is evaluated using an improved edge difference metric (iEDGE); for both, lower values indicate higher precision. Furthermore, the structural similarity index measure (SSIM), a widely adopted metric in computer vision, is used for auxiliary evaluation. The ablation studies verify the contributions of the ECHB, ETIE, QGSG, and the multihead attention strategy.

For quantitative metrics, RMSE is adopted for assessing spectral accuracy. It measures the deviation between the predicted image P and the ground truth F_P. The range of RMSE is $(0, +\infty)$, with values closer to 0 indicating higher spectral accuracy. The formula is calculated as follows:

$$\mathrm{RMSE}(P, F_P) = \sqrt{\mathbb{E}[(P - F_P)^2]}. \tag{11}$$

To quantify spatial accuracy, the Roberts operator extracts edge features from fused and ground truth images, and their differences are normalized to obtain the normalized edge difference value S_{nd}. This study adopts the improved EDGE (iEDGE) metric [33], [51], which robustly measures spatial accuracy by averaging absolute normalized edge differences within the top tenth percentile, yielding values in (0,1) where lower values indicate higher spatial accuracy. The computation process is as follows:

$$S_{\mathrm{nd}}(P, F_P) = \frac{\mathrm{Robert}(P) - \mathrm{Robert}(F_P)}{\mathrm{Robert}(P) + \mathrm{Robert}(F_P) + \varepsilon} \tag{12}$$

$$\mathrm{iEDGE}(P, F_P) = \mathbb{E}[|S_{\mathrm{nd}}(P, F_P)|]_{\mathrm{top10\%}} \tag{13}$$

where $\mathrm{Robert}(\cdot)$ represents the Roberts edge detection operator, $\mathbb{E}[\cdot]_{\mathrm{top10\%}}$ denotes the mean over the 90th-100th percentile range.

SSIM is a widely used metric for evaluating the perceptual similarity between a reference image and a processed image by coupling luminance and contrast comparisons via mean and covariance calculations of pixel values. The formula is defined as

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{\left(\mu_x^2 + \mu_y^2 + c_1\right)\left(\sigma_x^2 + \sigma_y^2 + c_2\right)} \tag{14}$$

where μ and σ represent the mean and variance, σ_{xy} is the covariance, and c_1 and c_2 are stability constants.

IV. Comparison and Analysis of Experimental Results

A. Qualitative Comparison

To evaluate the performance of ArtFusion across diverse and complex conditions, local magnified views from four challenging scenarios were selected for comparative analysis. The first scenario encompasses areas undergoing rapid phenological changes, covering both agricultural and natural land cover types, where accurate prediction is hampered by rapid spectral transitions. The second scenario focuses on drastic morphological changes, including seasonal fluctuations of the Poyang Lake wetland and the large-scale flooding event in the Southern Taihang Mountains Area, where significant alterations in land-cover patterns pose considerable challenges for reconstruction. The third scenario involves strongly heterogeneous landscapes, where heterogeneous information and small-scale changes are difficult to restore given the prediction-phase coarse imagery. The fourth scenario tests frequent cloud contamination, evaluating the capability to exploit available reliable information effectively.

1) Rapid Phenological Changes: As shown in Fig. 3, both study areas exhibit rapid phenological dynamics. Fig. 3(a) and (b) presents the reconstruction results for fragmented croplands and mountainous natural vegetation surrounding Poyang Lake. Observations indicate that MLFF-GAN, RealFusion, and FSDFormer produce comparatively blurry outputs, attributable to the limited reference imagery available to capture such dramatic and heterogeneous phenological changes. While C-ROBOT recovers textures effectively, its spectral accuracy remains compromised. In contrast, ArtFusion achieves the highest fidelity. Fig. 3(c)–(e) displays results for the Western Kansas Agricultural Zone. C-ROBOT exhibits limitations in both spectral and spatial restoration, possibly due to phenological discrepancies between adjacent reference images. In contrast, MLFF-GAN, RealFusion, FSDFormer, and ArtFusion all achieve relatively accurate spectral reconstruction. However, MLFF-GAN and FSDFormer suffer from noticeable checkerboard artifacts and blurred boundaries. RealFusion fails to accurately reconstruct certain farmland geometries (e.g., insufficiently circular fields), whereas ArtFusion provides the closest approximation to the ground truth in both spectral consistency and spatial texture.

2) Drastic Morphological Changes: Fig. 4 presents local results for scenarios involving drastic morphological changes, including the seasonal wetlands of Poyang Lake and flood-affected croplands in the South Taihang Mountains Area. The former is characterized by recurrent cycles of lake-bed exposure and inundation, whereas the latter involves large-scale cropland submergence caused by an extreme rainfall event. This scenario is particularly challenging as it violates the stationarity assumption of land-cover types and structures. MLFF-GAN, C-ROBOT, FSDFormer, and RealFusion all exhibit noticeable artifacts. MLFF-GAN and FSDFormer exhibit prominent checkerboard artifacts, a manifestation of the model's failure to accurately reconstruct complex structural shapes. C-ROBOT shows distinct blocky artifacts due to limited regression performance despite the use of long-term

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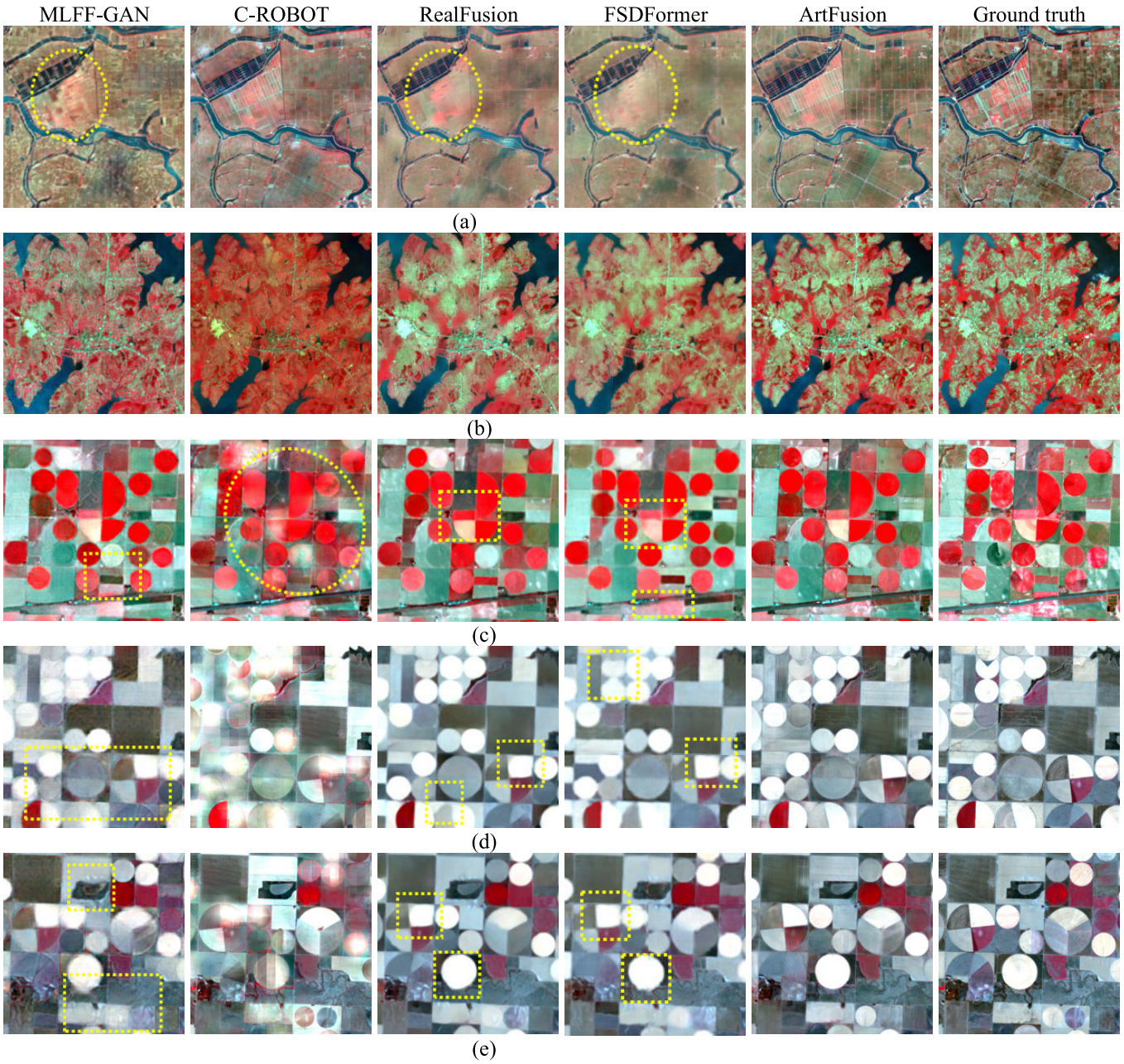


Fig. 3. Comparison of experimental results in regions with rapid phenological changes. (a) Fragmented cropland subregion in the Poyang Lake Wetland (Prediction phase: February 16, 2016). (b) Mountainous subregion in the Poyang Lake Wetland (Prediction phase: September 30, 2017). (c) Subregion in the Western Kansas Agricultural Zone (Prediction phase: June 27, 2020). (d) Subregion in the Western Kansas Agricultural Zone (Prediction phase: November 5, 2021). (e) Subregion in the Western Kansas Agricultural Zone (Prediction phase: November 5, 2021).

sequences. RealFusion performs better due to the integration of annual NDVI. However, its results remain blurred, featuring artificial striation artifacts caused by the inherent ill-posedness of the task. In contrast, ArtFusion generates highly realistic predictions with substantially higher structural and spectral fidelity, closely matching the ground truth even under these challenging conditions. The superiority of ArtFusion is particularly evident in Fig. 4(g), where agricultural fields were inundated by an unprecedented flood event. ArtFusion accurately reconstructs the spatial extent and boundaries of the flooded areas, while the competing methods either miss significant portions of the inundation or produce severe structural distortions. Although some spectral deviations remain in the reconstructed water bodies, this is expected because no flood-related samples were included in the training set. Nevertheless, ArtFusion still delivers markedly more accurate results than all four competing methods, demonstrating strong robustness and generalization capability when confronted with previously unseen large-scale land-surface changes.

3) Highly Heterogeneous Landscapes: Reconstructing spatial information in highly heterogeneous areas is a severely ill-posed problem owing to the substantial resolution gap,

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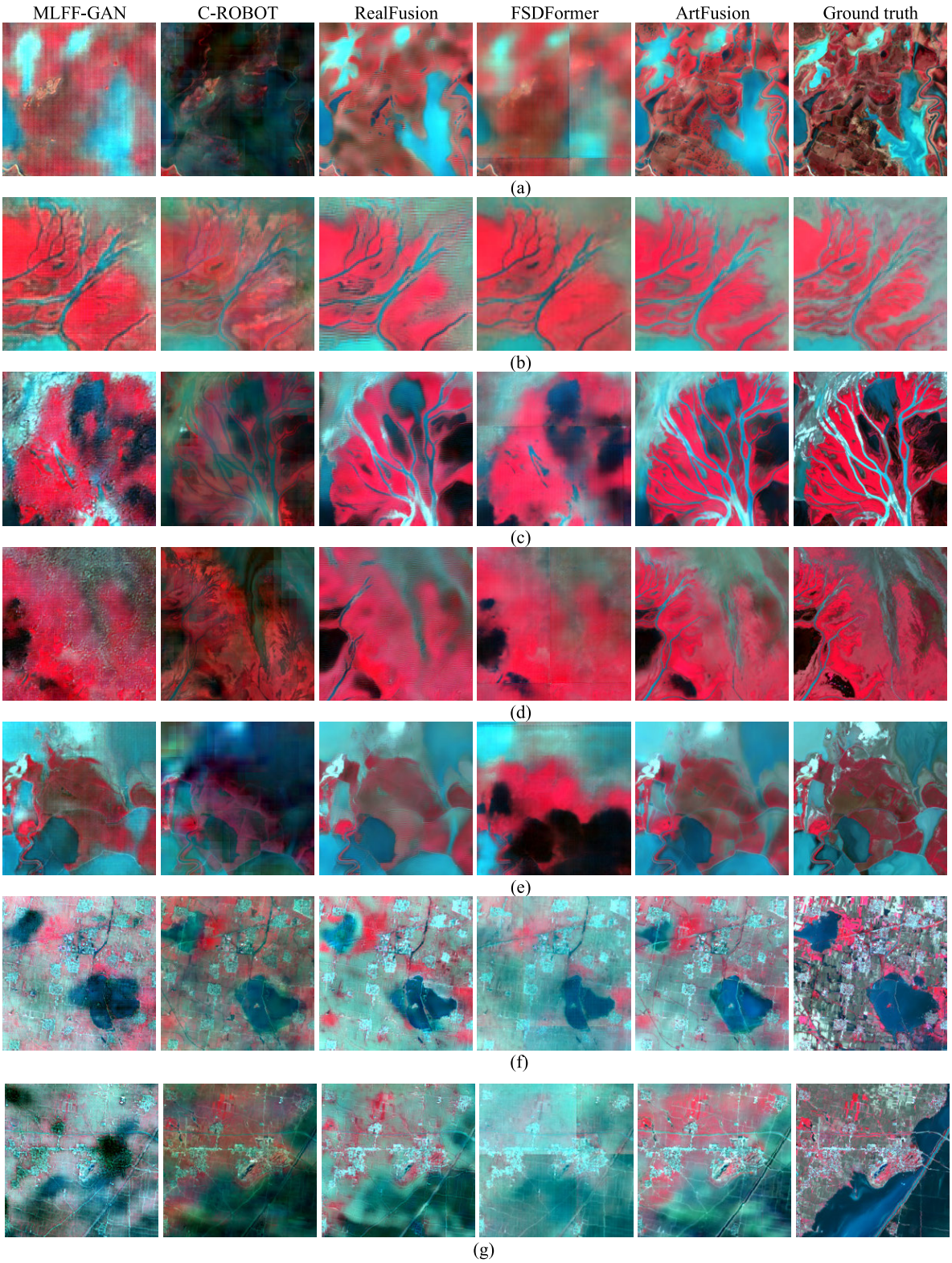


Fig. 4. Comparison of experimental results in regions with land surface morphology changes. (a) Seasonal wetland subregion in the Poyang Lake Wetland (Prediction phase: March 14, 2014). (b) Seasonal wetland subregion in the Poyang Lake Wetland (Prediction phase: December 16, 2016). (c) Seasonal wetland subregion in the Poyang Lake Wetland (Prediction phase: October 3, 2018). (d) Seasonal wetland subregion in the Poyang Lake Wetland (Prediction phase: October 3, 2018). (e) Seasonal wetland subregion in the Poyang Lake Wetland (Prediction phase: September 20, 2019). (f) Flood inundation subregion in the Southern Taihang Mountains Area (Prediction phase: September 30, 2021). (g) Flood inundation subregion in the Southern Taihang Mountains Area (Prediction phase: September 30, 2021).

making multitemporal auxiliary data indispensable for supplying critical spatial prior information. Fig. 5(a) shows small-scale surface dynamics associated with urban construction activities in Nanchang. Among the evaluated methods,

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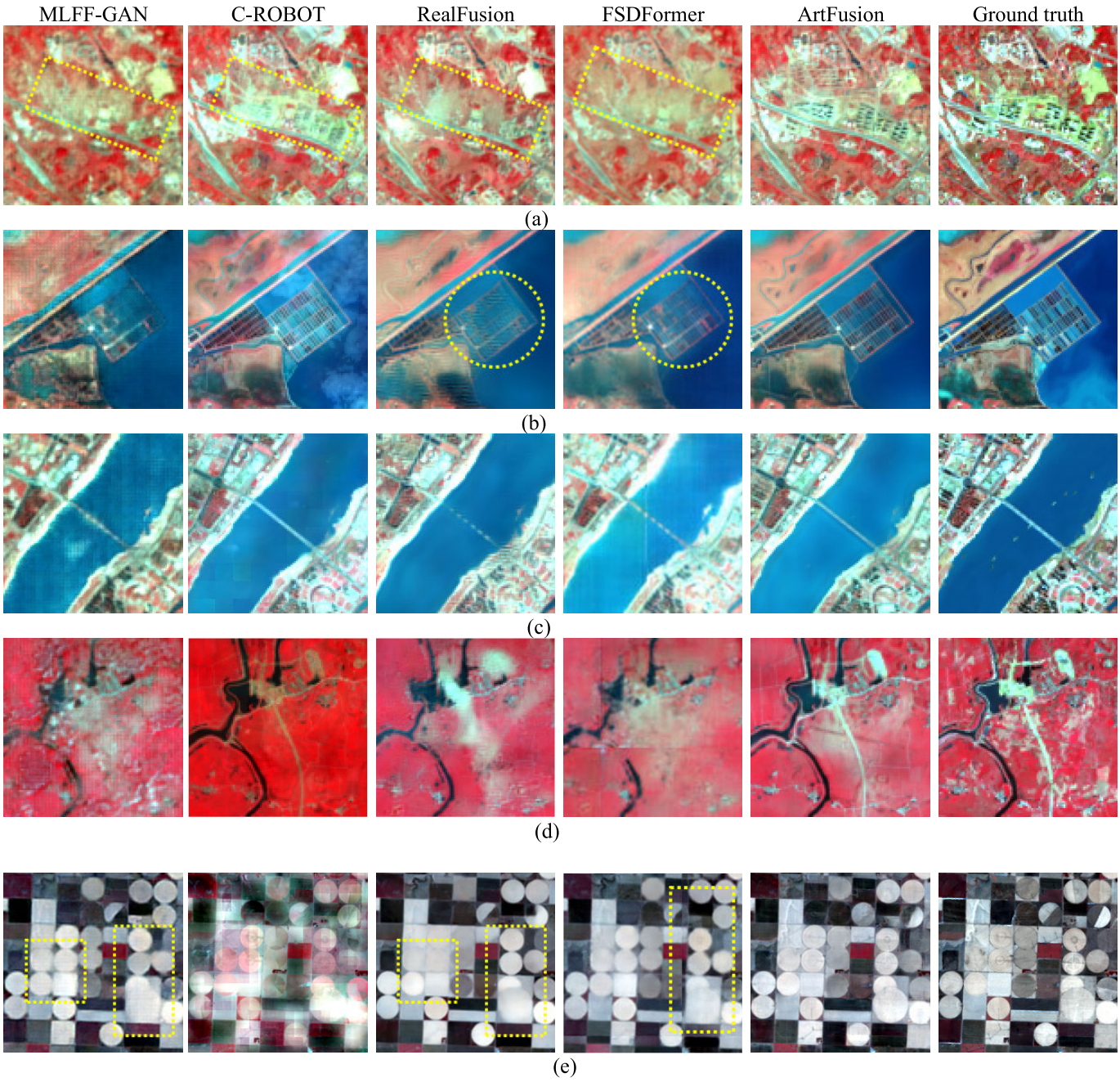


Fig. 5. Comparison of experimental results in highly heterogeneous landscape regions. (a) Urban subregion in the Poyang Lake Wetland (Prediction phase: February 16, 2016). (b) Aquaculture pond subregion in the Poyang Lake Wetland (Prediction phase: February 16, 2016). (c) Urban bridge subregion in the Poyang Lake Wetland (Prediction phase: February 16, 2016). (d) Urban road subregion in the Poyang Lake Wetland (Prediction phase: October 3, 2018). (e) Subregion in the Western Kansas Agricultural Zone (Prediction phase: January 3, 2020).

only C-ROBOT and ArtFusion achieve accurate reconstruction of such dynamics, with ArtFusion yielding finer-grained and more reliable building texture details. Fig. 5(b) displays the reconstruction results for aquaculture ponds, where ArtFusion demonstrates superior performance in fine-detail restoration. For linear features such as bridges and roads [Fig. 5(c) and (d)], which are typically only ~2 pixels wide, MLFF-GAN, FSDFormer, and RealFusion fail to capture these elements, as such features are absent from their constrained reference datasets. Finally, for the texture-dense croplands in the Western Kansas Agricultural Region [Fig. 5(e)], ArtFusion emerges as the sole method capable of faithfully reconstructing the internal texture details of crop canopies.

4) Frequent Cloud Contamination: Cloud contamination of auxiliary images exerts a pronounced detrimental effect on STF accuracy, with single-reference-pair fusion methods being particularly vulnerable to such interference. As illustrated in Fig. 6, when temporally adjacent input reference images are affected by cloud cover, ArtFusion stands out as the sole method capable of reliably reconstructing the underlying

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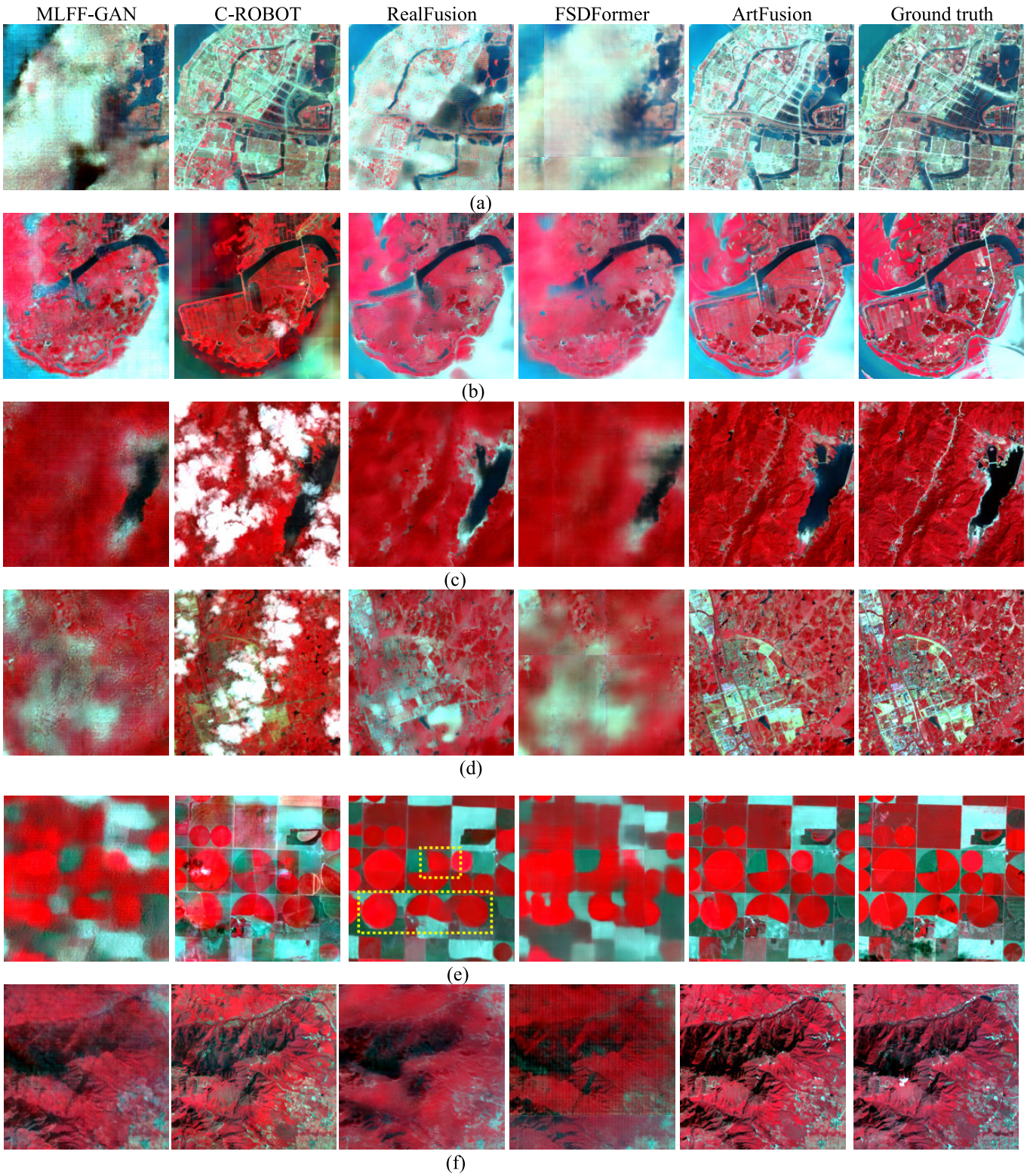


Fig. 6. Comparison of experimental results in cloud-prone scenarios. (a) Urban subregion in the Poyang Lake Wetland (Prediction phase: March 14, 2014). (b) Wetland subregion in the Poyang Lake Wetland (Prediction phase: October 3, 2018). (c) Mountainous wild subregion in the Poyang Lake Wetland (Prediction phase: October 3, 2018). (d) Urban subregion in the Poyang Lake Wetland (Prediction phase: October 3, 2018). (e) Subregion in the Western Kansas Agricultural Zone (Prediction phase: August 28, 2019). (f) Subregion in the Southern Taihang Mountains Area. (Prediction phase: September 14, 2015).

land-surface information to achieve close alignment with the ground truth. In contrast, MLFF-GAN and FSDFormer fail to retrieve spatially coherent land-surface details beneath cloud-covered regions. While C-ROBOT incorporates long-term imagery, its reliance on temporally adjacent Landsat constraints, intended to enhance robustness, unintentionally

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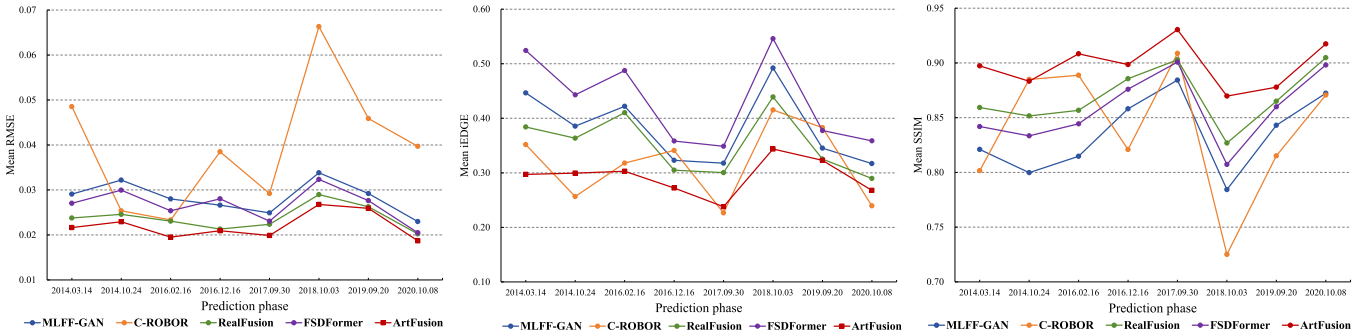


Fig. 7. Accuracy performance across all experiments in the Poyang Lake Wetland. (a) Mean RMSE of six bands. (b) Mean iEDGE of six bands. (c) Mean SSIM of six bands.

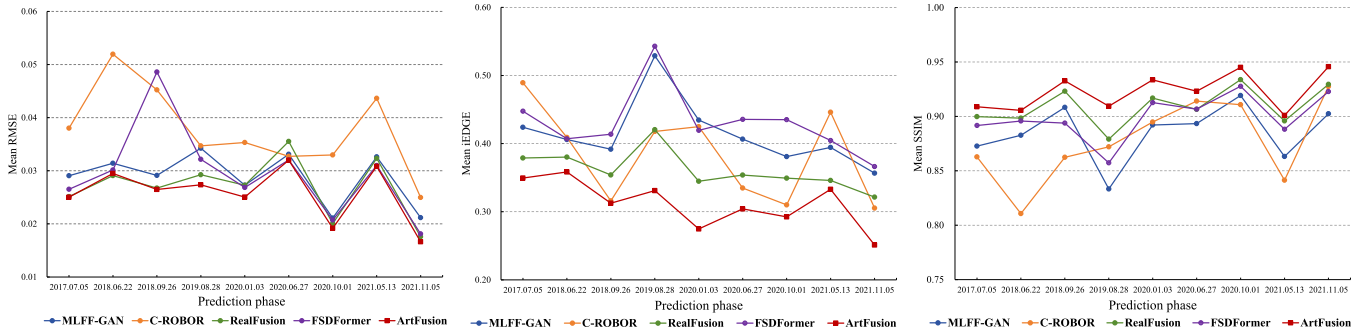


Fig. 8. Accuracy performance across all experiments in the Western Kansas Agricultural Zone. (a) Mean RMSE of six bands. (b) Mean iEDGE of six bands. (c) Mean SSIM of six bands.

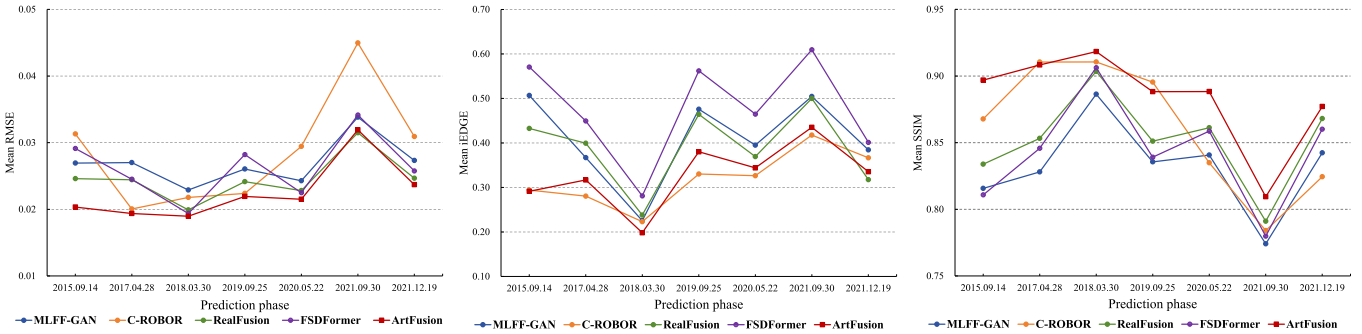


Fig. 9. Accuracy performance across all experiments in the Southern Taihang Mountains Area. (a) Mean RMSE of six bands. (b) Mean iEDGE of six bands. (c) Mean SSIM of six bands.

introduces cloud-related uncertainties, leading to residual spectral distortions and cloud-shaped artifacts. RealFusion leverages annual NDVI data to alleviate cloud-related interference, but it exhibits limited sensitivity in nonvegetated regions, ultimately leading to the generation of blurred surface textures. ArtFusion demonstrates a clear advantage by effectively distilling reliable features from the multitemporal stack, even under conditions of severe cloud contamination.

B. Quantitative Comparison

Figs. 7–9 present the accuracy curves of individual experiments in the three study regions, while Tables II–IV summarize the average quantitative results across all experiments, with the best values highlighted in bold. The results show that ArtFusion achieves the best performance for nearly all metrics. Compared with the strongest competing method, RealFusion, ArtFusion reduces the average RMSE by 7.6%, 4.4%, and 8.5% in the Poyang Lake Wetland, Western Kansas Agricultural Zone, and Southern Taihang Mountains Area, respectively. Similarly, the average iEDGE is reduced by 16.8%, 13.6%, and 15.4%, indicating superior preservation of both spectral fidelity and fine-scale spatial structures.

Overall, ArtFusion consistently outperforms the advanced comparative methods across all three study regions and a wide range of challenging scenarios. The quantitative results confirm its superior reconstruction

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TABLE II

AVERAGE QUANTITATIVE EVALUATION RESULTS ACROSS ALL EXPERIMENTS IN THE POYANG LAKE WETLAND

Method	Metric	Blue	Green	Red	NIR	SWIR1	SWIR2	Mean
MLFF-GAN	RMSE↓	0.0138	0.0160	0.0212	0.0445	0.0422	0.0324	0.0284
	iEDGE↓	0.3845	0.4074	0.3926	0.3755	0.3631	0.3638	0.3812
	SSIM↑	0.9361	0.9281	0.8788	0.7476	0.7412	0.7762	0.8347
C-ROBOT	RMSE↓	0.0257	0.0274	0.0320	0.0628	0.0513	0.0384	0.0396
	iEDGE↓	**0.3255**	0.3311	0.3241	0.3045	0.3051	0.3082	0.3164
	SSIM↑	0.9212	0.9200	0.8772	0.7556	0.7640	0.7986	0.8395
RealFusion	RMSE↓	0.0115	0.0137	0.0179	0.0360	0.0359	0.0280	0.0238
	iEDGE↓	0.3664	0.3806	0.3576	0.3317	0.3395	0.3371	0.3521
	SSIM↑	0.9497	0.9420	0.9030	0.7966	0.7939	0.8293	0.8691
FSDFormer	RMSE↓	0.0131	0.0152	0.0200	0.0419	0.0395	0.0307	0.0267
	iEDGE↓	0.4602	0.4718	0.4592	0.3951	0.3919	0.4050	0.4305
	SSIM↑	0.9493	0.9395	0.8978	0.7761	0.7731	0.8104	0.8577
ArtFusion	RMSE↓	**0.0104**	**0.0128**	**0.0166**	**0.0340**	**0.0329**	**0.0255**	**0.0220**
	iEDGE↓	0.3293	**0.3301**	**0.3152**	**0.2585**	**0.2591**	**0.2663**	**0.2931**
	SSIM↑	**0.9611**	**0.9531**	**0.9232**	**0.8389**	**0.8413**	**0.8693**	**0.8978**

TABLE III

AVERAGE QUANTITATIVE EVALUATION RESULTS ACROSS ALL EXPERIMENTS IN THE WESTERN KANSAS AGRICULTURAL ZONE

Method	Metric	Blue	Green	Red	NIR	SWIR1	SWIR2	Mean
MLFF-GAN	RMSE↓	0.0137	0.0176	0.0240	0.0353	0.0426	0.0395	0.0288
	iEDGE↓	0.4079	0.4009	0.3982	0.4490	0.4287	0.3976	0.4137
	SSIM↑	0.9482	0.9336	0.8983	0.8573	0.8329	0.8412	0.8852
C-ROBOT	RMSE↓	0.0241	0.0268	0.0341	0.0427	0.0520	0.0467	0.0377
	iEDGE↓	0.3800	0.3649	0.3838	0.3970	0.4025	0.3739	0.3837
	SSIM↑	0.9252	0.9195	0.8816	0.8566	0.8344	0.8468	0.8774
RealFusion	RMSE↓	0.0132	0.0160	0.0213	0.0321	0.0406	0.0389	0.0270
	iEDGE↓	0.3542	0.3424	0.3354	0.3933	0.3919	0.3486	0.3610
	SSIM↑	0.9587	0.9498	0.9237	0.8859	0.8661	0.8710	0.9092
FSDFormer	RMSE↓	0.0167	0.0197	0.0254	0.0343	0.0419	0.0395	0.0296
	iEDGE↓	0.4276	0.4165	0.4095	0.4790	0.4418	0.4073	0.4303
	SSIM↑	0.9509	0.9401	0.9103	0.8796	0.8552	0.8616	0.8996
ArtFusion	RMSE↓	**0.0121**	**0.0155**	**0.0209**	**0.0315**	**0.0386**	**0.0361**	**0.0258**
	iEDGE↓	**0.3145**	**0.2941**	**0.2894**	**0.3482**	**0.3223**	**0.3026**	**0.3118**
	SSIM↑	**0.9647**	**0.9574**	**0.9346**	**0.9008**	**0.8880**	**0.8910**	**0.9228**

accuracy, robustness, and generalization capability for complex STF tasks.

C. Ablation Study

To evaluate the contribution of each proposed component, ablation experiments were conducted with four variants: 1) replacing the ECHB with CNN residual blocks to assess feature extraction capability; 2) removing the ETIE to evaluate the effect of temporal information integration; 3) removing the QGSG to examine the contribution of spatial attention; and 4) disabling the multihead attention (Mhead) strategy to verify the effectiveness of parallel subspace modeling. The quantitative results are summarized in Tables V–VII, with the best results highlighted in bold. The full ArtFusion model achieves the highest accuracy across all four metrics, demonstrating that ECHB, ETIE, QGSG, and Mhead collaboratively contribute to improved feature representation, temporal guidance, spatial refinement, and fusion performance.

TABLE IV

AVERAGE QUANTITATIVE EVALUATION RESULTS ACROSS ALL EXPERIMENTS IN THE SOUTHERN TAIHANG MOUNTAINS AREA

Method	Metric	Blue	Green	Red	NIR	SWIR1	SWIR2	Mean
MLFF-GAN	RMSE↓	0.0150	0.0172	0.0223	0.0363	0.0365	0.0341	0.0269
	iEDGE↓	0.4029	0.4109	0.4170	0.4117	0.4106	0.3977	0.4085
	SSIM↑	0.8985	0.8943	0.8406	0.7223	0.7711	0.7675	0.8157
C-ROBOT	RMSE↓	0.0204	0.0207	0.0251	0.0375	0.0352	0.0333	0.0287
	iEDGE↓	**0.2868**	**0.3038**	**0.3328**	0.3309	0.3266	0.3380	**0.3198**
	SSIM↑	0.9213	0.9241	0.8775	0.8038	0.8203	0.8197	0.8611
RealFusion	RMSE↓	0.0129	0.0154	0.0205	0.0342	0.0328	0.0316	0.0246
	iEDGE↓	0.3835	0.3922	0.3923	0.4052	0.3767	0.3828	0.3888
	SSIM↑	0.9334	0.9214	0.8744	0.7789	0.7996	0.8024	0.8517
FSDFormer	RMSE↓	0.0169	0.0192	0.0228	0.0338	0.0327	0.0319	0.0262
	iEDGE↓	0.4593	0.4734	0.4951	0.4925	0.4611	0.4805	0.4770
	SSIM↑	0.9154	0.9085	0.8621	0.7778	0.7962	0.7975	0.8429
ArtFusion	RMSE↓	**0.0120**	**0.0141**	**0.0190**	**0.0311**	**0.0300**	**0.0290**	**0.0225**
	iEDGE↓	0.3169	0.3317	0.3496	**0.3260**	**0.3169**	**0.3314**	0.3288
	SSIM↑	**0.9453**	**0.9364**	**0.8980**	**0.8289**	**0.8475**	**0.8468**	**0.8838**

TABLE V

AVERAGE QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS IN THE POYANG LAKE WETLAND

Modules				Metrics		
ECHB	ETIE	QGSG	Mhead	Mean RMSE↓	Mean iEDGE↓	Mean SSIM↑
×	√	√	√	0.0227	0.3007	0.8944
√	×	√	√	0.0225	0.2990	0.8941
√	√	×	√	0.0221	0.2980	0.8960
√	√	√	×	0.0221	0.2966	0.8971
√	√	√	√	**0.0220**	**0.2931**	**0.8978**

TABLE VI

AVERAGE QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS IN THE WESTERN KANSAS AGRICULTURAL ZONE

Modules				Metrics		
ECHB	ETIE	QGSG	Mhead	Mean RMSE↓	Mean iEDGE↓	Mean SSIM↑
×	√	√	√	0.0261	0.3196	0.9195
√	×	√	√	0.0261	0.3164	0.9210
√	√	×	√	0.0263	0.3192	0.9195
√	√	√	×	0.0261	0.3138	0.9226
√	√	√	√	**0.0258**	**0.3118**	**0.9228**

TABLE VII

AVERAGE QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS IN THE SOUTHERN TAIHANG MOUNTAINS AREA

Modules				Metrics		
ECHB	ETIE	QGSG	Mhead	Mean RMSE↓	Mean iEDGE↓	Mean SSIM↑
×	√	√	√	0.0244	0.3366	0.8736
√	×	√	√	0.0226	0.3393	0.8836
√	√	×	√	0.0235	0.3347	0.8810
√	√	√	×	0.0228	**0.3268**	0.8822
√	√	√	√	**0.0225**	0.3288	**0.8838**

V. DISCUSSION

A. Comparison of Fusion Frameworks

Fig. 10 illustrates the fusion frameworks of representative STF methods from the perspective of reference-image utilization. Existing STF methods generally suffer from a common intrinsic limitation: the inability to effectively exploit the rich

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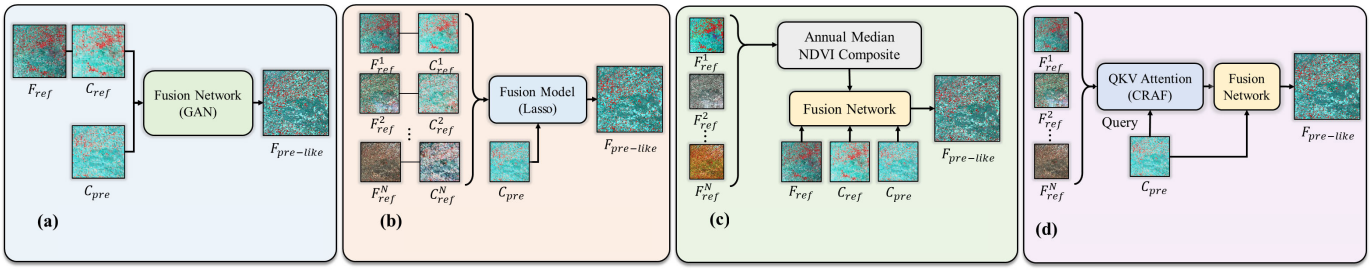


Fig. 10. Schematic illustrations of the fusion frameworks for four representative STF methods. (a) MLFF-GAN: single-reference-pair fusion. (b) ROBOT: multiple-reference-pair fusion. (c) RealFusion: composited-NDVI-assisted fusion. (d) ArtFusion: multiple-reference fusion.

spatiotemporal information contained in long-term historical image archives.

Single-reference-pair methods, represented by MLFF-GAN, currently dominate deep learning-based STF research. Although advanced neural network architectures can improve feature representation, the reconstruction process remains fundamentally constrained by the limited information provided by one or two reference image pairs. Consequently, fusion accuracy is highly sensitive to the quality and representativeness of the selected references. When the reference images are affected by cloud contamination, phenological inconsistency, or abrupt land-surface changes, the available information becomes insufficient to adequately constrain the highly ill-posed STF problem.

ROBOT partially alleviates this limitation by utilizing multiple image pairs. However, its performance depends on the assumption of consistent information changes among reference observations. In practice, the relevance of different reference images varies substantially due to sensor inconsistencies and image-quality variations. Consequently, simply increasing the number of reference images does not necessarily improve accuracy and may even introduce additional uncertainty. RealFusion adopts annual NDVI composites as auxiliary information to compensate for missing details. While this strategy improves robustness under cloud contamination, the transformation of multispectral observations into a single-band NDVI representation inevitably discards substantial spatiotemporal information, limiting its ability to characterize complex changes.

Accordingly, the core bottleneck of STF lies not in elaborate network design, but in efficient exploitation of massive historical observations. ArtFusion was specifically developed to address this challenge. By introducing a flexible multireference fusion framework, ArtFusion enables the direct utilization of an arbitrary number of reference images and fully exploits the rich spatiotemporal information contained in historical archives. The ETIE module explicitly incorporates temporal metadata, while the MCAF module adaptively evaluates the relevance of different observations and selectively aggregates useful information under the guidance of prediction-phase MODIS data. Consequently, ArtFusion can automatically emphasize high-quality and highly relevant references while suppressing less informative observations.

From this perspective, the core contribution of ArtFusion lies not in the introduction of individual network components but in establishing a new STF framework capable of effectively utilizing arbitrary numbers of reference images and extensive historical archives. This framework fundamentally differs from existing STF paradigms and provides a promising pathway for overcoming the information-utilization bottleneck that has long constrained STF performance.

B. Sensitivity to the Number of Reference Images

Previous research has shown that temporally proximal reference images can substantially improve STF performance [52]. However, due to inherent land-surface dynamics, fusion accuracy does not increase indefinitely with the number of inputs. Excessive reference images may introduce spectral noise or contradictory land-cover information, potentially leading to performance saturation or degradation [44]. To investigate this behavior, sensitivity experiments were conducted by progressively increasing the number of reference images from 1 to 15 for the Poyang Lake Wetland and from 1 to 11 for the Western Kansas Agricultural Zone, with inputs ranked by temporal proximity to the prediction phase. Mean RMSE and iEDGE across six bands were adopted to evaluate spectral and spatial accuracy, respectively. The performance of C-ROBOT, MLFF-GAN, RealFusion, and FSDFormer was also included for comparison, with the latter three serving as fixed baselines.

As illustrated in Fig. 11, ArtFusion exhibits a clear saturation characteristic as the number of reference images increases. Both spectral and spatial accuracies improve rapidly when a small number of informative reference images are introduced, followed by a stable plateau after the optimal range is reached. Spectral accuracy (RMSE) remains remarkably stable with negligible degradation across all four evaluation scenarios, whereas spatial fidelity (iEDGE) shows a slight decline after reaching its optimum in three of the four evaluation scenarios. This behavior suggests that redundant reference images may dilute the attention assigned to the most relevant observations, resulting in mild over-smoothing that compromises fine-grained spatial textures. Overall, when temporally adjacent images are progressively incorporated, optimal performance is generally achieved with 2–8 reference images. Beyond this range, additional inputs contribute limited new information and mainly lead to performance saturation rather than further improvement. These observations also provide practical guidance for different applications. For applications

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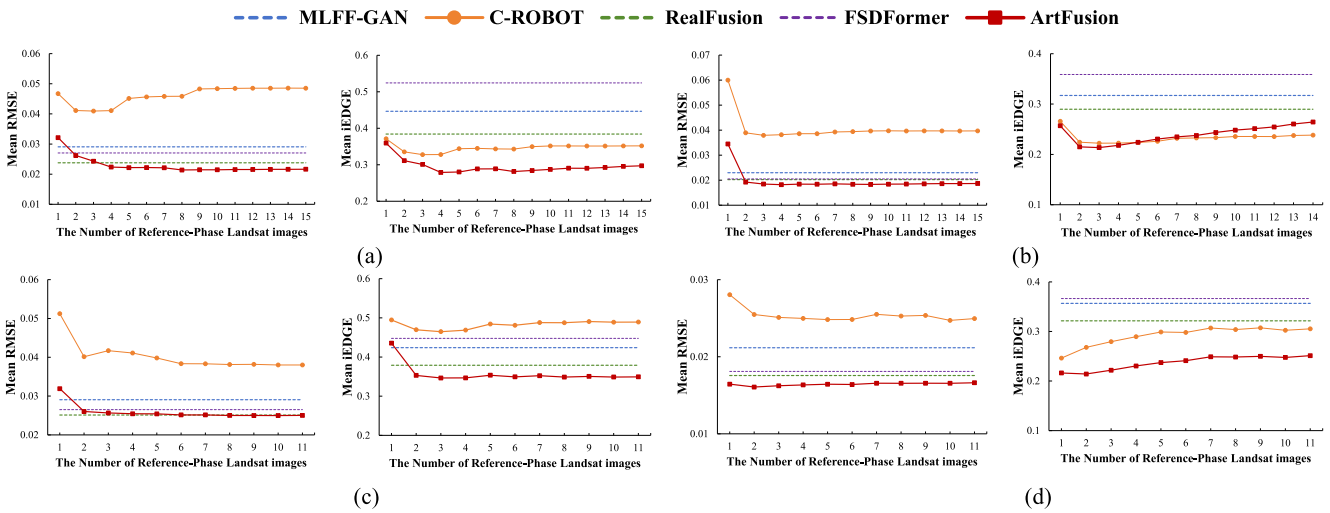


Fig. 11. Variations in fusion accuracy of STF relative to the number of input fine reference images. (a) Poyang Lake Wetland (Prediction phase: March 14, 2014). (b) Poyang Lake Wetland (Prediction phase: October 8, 2020). (c) Western Kansas Agricultural Zone (Prediction phase: July 5, 2017). (d) Western Kansas Agricultural Zone (Prediction phase: November 5, 2021).

emphasizing spectral consistency, such as phenological monitoring, a relatively larger number of reference images can be adopted to improve robustness against cloud contamination and noise. In contrast, for applications requiring high spatial fidelity, such as land-cover boundary extraction, preliminary reference-image screening is recommended to prioritize high-quality reference images with strong temporal and phenological relevance to the target phase.

A cross-comparison further highlights the robustness of ArtFusion. Although C-ROBOT also exploits multitemporal sequences, its spectral and spatial accuracies deteriorate noticeably as the number of reference images increases. For example, in the Poyang Lake Wetland experiment (Prediction phase: 2014.03.14), C-ROBOT reached its optimum with four reference images and declined sharply thereafter. In contrast, ArtFusion maintained stable performance after saturation and consistently outperformed C-ROBOT, MLFF-GAN, RealFusion, and FSDFormer when more than four reference images were used. These results demonstrate that ArtFusion can effectively exploit informative multitemporal observations while suppressing redundant interference, resulting in superior reconstruction reliability and accuracy.

C. Computational Efficiency and Complexity

The computational complexity of the proposed ArtFusion model was evaluated in terms of trainable parameters and computational overhead. For a standardized comparison, all models were tested using an input size of 400×400 and a batch size of 1, except for FSDFormer, which was evaluated with an input size of 256×256 due to architectural constraints. During the training phase, ArtFusion was configured with nine reference Landsat images. To provide a comprehensive perspective, the benchmarks included not only recent SOTA methods (RealFusion and MLFF-GAN) but also earlier lightweight architectures such as the enhanced deep convolutional STF network (EDCSTFN) [53] and GAN-based STF model (GAN-STFM) [28].

TABLE VIII

COMPARISON OF MODEL COMPLEXITY ACROSS DIFFERENT METHODS. THE MOST LIGHTWEIGHT RESULTS ARE SHOWN IN BOLD

Model	Parameters (M)	FLOPs (G)
EDCSTFN	0.28	**45.4**
GAN-STFM	4.24	93.6
MLFF-GAN	8.70	49.2
RealFusion	21.65	86.0
FSDFormer	3.69	24.8*
ArtFusion+SIQRN-lite	4.04	148.02
ArtFusion	**0.19**	139.68

As summarized in Table VIII, ArtFusion boasts the most compact architecture among all evaluated models. Excluding the SIQRN-lite module, its core network contains merely 0.19-M parameters. Even with the integration of SIQRN-lite, the total parameter count remains lower than that of GAN-STFM and far below that of the SOTA MLFF-GAN and RealFusion models. This compactness underscores the efficiency of the ECHB and MCAF modules in deep feature extraction and multireference fusion without redundant overhead.

Regarding computational intensity, ArtFusion exhibits higher FLOPs compared to single-reference-pair fusion methods. This increase is an expected consequence of the simultaneous processing of multiple fine reference streams to achieve superior reconstruction. However, this computational demand is highly scalable and manageable in practical deployments. Leveraging the saturation effect identified in Section V-A, where optimal performance is attained with 2–8 reference images, the computational footprint can be dynamically optimized. By selecting an appropriate number of references based on available hardware constraints, users can substantially reduce FLOPs while maintaining a negligible impact on reconstruction accuracy. This flexibility ensures that ArtFusion can be effectively deployed across diverse computing environments, from high-performance workstations to resource-constrained scenarios.

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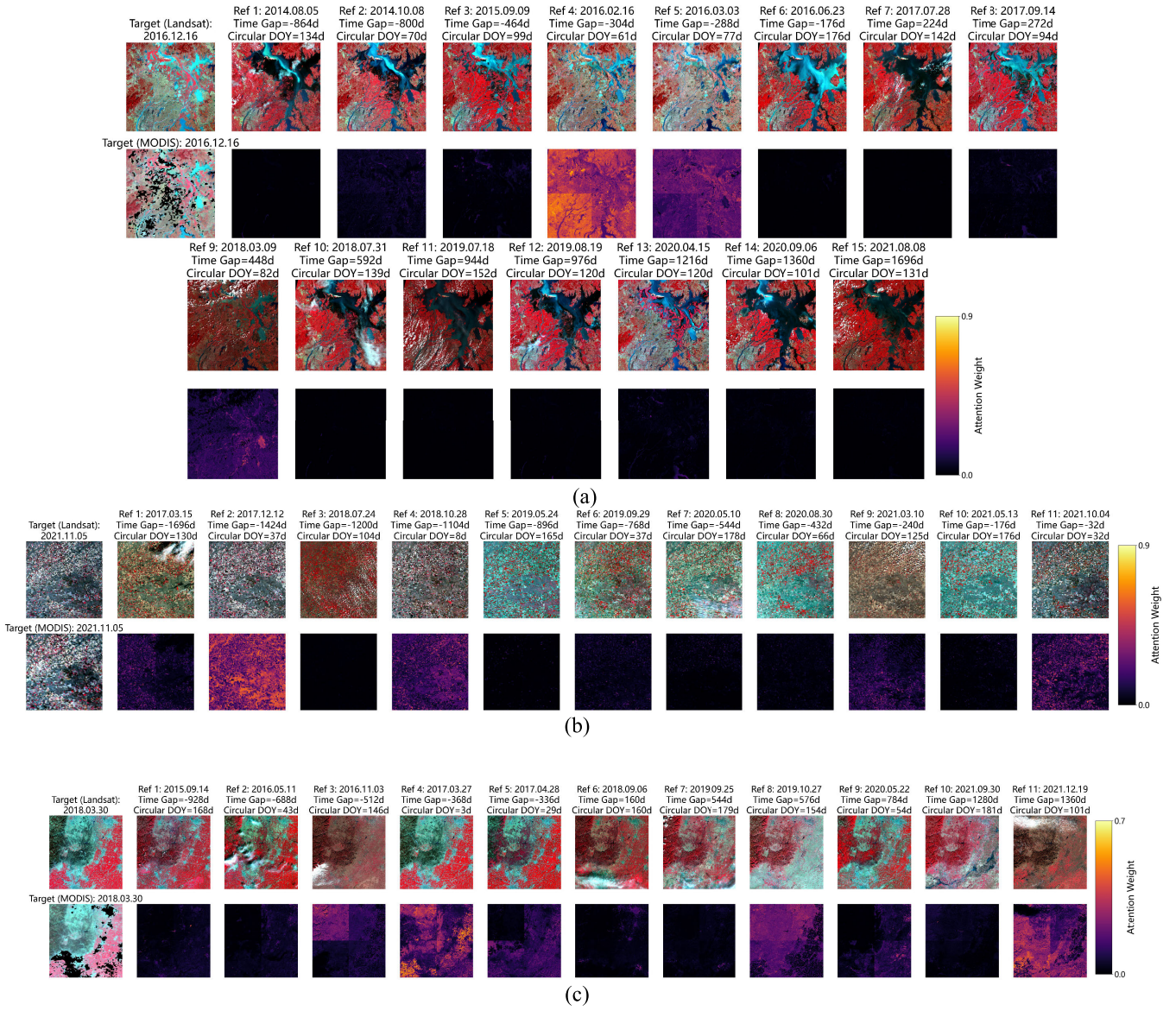


Fig. 12. Visualization of reference-image attention weights in representative experiments across the three study areas. (a) Poyang Lake Wetland (Prediction phase: December 16, 2016). (b) Western Kansas Agricultural Zone (Prediction phase: November 5, 2021). (c) Southern Taihang Mountains Area (Prediction phase: March 30, 2018).

D. Visualization Analysis of Reference Image Weights

To investigate the key factors governing the weight allocation of reference images in ArtFusion, a visualization analysis was conducted across the three study areas. Representative examples are presented in Fig. 12, while the scatter distributions between attention weights and temporal gaps, as well as between attention weights and circular DOY gaps, for all experiments in the three study areas are shown in Figs. 13 and 14, respectively.

The results indicate that the attention weights assigned to reference images exhibit pronounced spatiotemporal variability and are jointly governed by image quality, temporal proximity, and phenological similarity. As illustrated in Fig. 12, cloud-contaminated images consistently receive lower weights, indicating that the model can automatically identify low-quality observations and suppress their influence during information fusion. Fig. 13 further shows that reference images acquired closer to the target phase generally receive higher weights. This tendency is particularly evident in the Poyang Lake Wetland and Southern Taihang Mountains Area, where rapid urbanization and intensive human activities result in persistent land-surface changes. In terms of phenological similarity, Figs. 12 and 14 reveal that reference images acquired under similar seasonal conditions tend to receive higher weights. For example, in the Poyang Lake Wetland, dry-season observations are generally assigned greater importance than those acquired during the wet season. However, the weighting strategy is not governed by any single factor. Reference images with moderate spectral differences may still receive relatively high weights when the corresponding land-surface changes follow predictable temporal trajectories that can be effectively modeled by the network.

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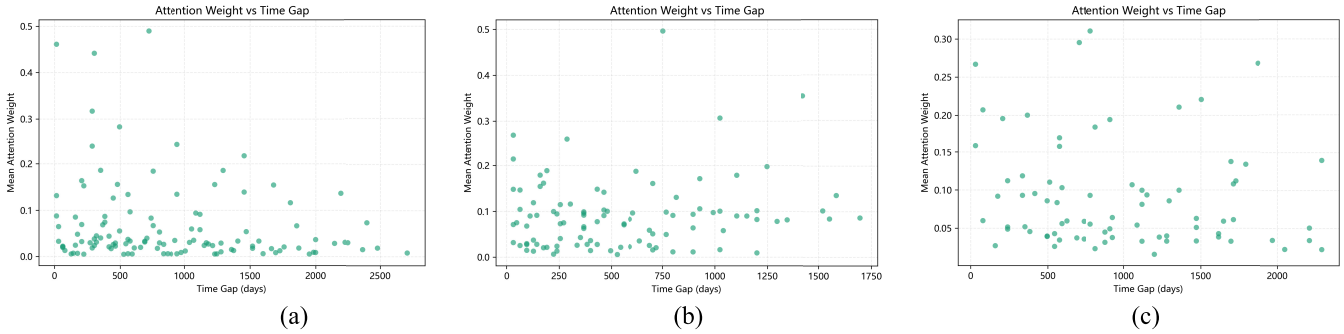


Fig. 13. Scatter plots of attention weights versus temporal gaps for all experiments in the three study areas. (a) Poyang Lake Wetland. (b) Western Kansas Agricultural Zone. (c) Southern Taihang Mountains Area.

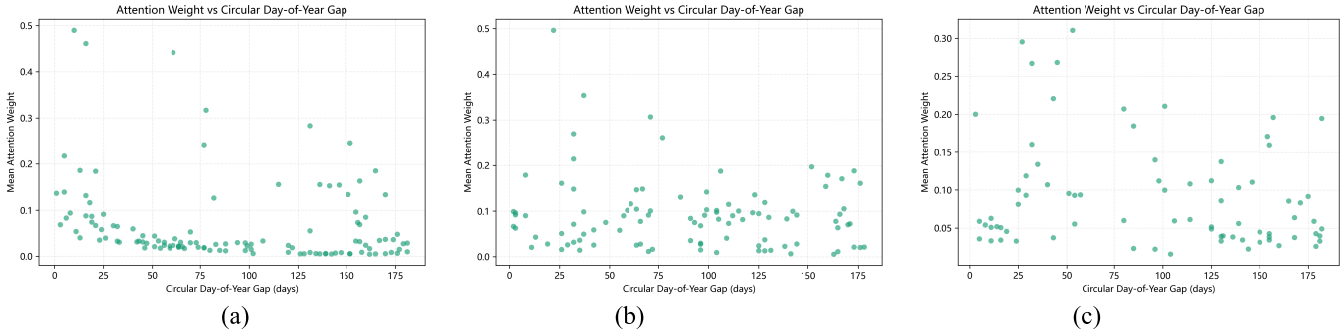


Fig. 14. Scatter plots of attention weights versus circular DOY gaps for all experiments in the three study areas. (a) Poyang Lake Wetland. (b) Western Kansas Agricultural Zone. (c) Southern Taihang Mountains Area.

Overall, the attention weights in ArtFusion originate from the spatiotemporal similarity between the reference and target images. This similarity is jointly determined by image quality, temporal proximity, and phenological similarity rather than any single factor. Such an adaptive weighting mechanism enables robust reference-image selection and provides a key foundation for generating high-quality fusion results.

E. Reference Image Selection Strategy

Based on the analyses in Sections V-B–V-D, different reference images contribute unequally to the fusion results, and a small number of high-value images usually provide the majority of effective information. Therefore, increasing the number of reference images does not necessarily lead to better performance; excessive redundant inputs may increase computational costs and cause performance saturation. Although ArtFusion supports an arbitrary number of reference image inputs, selecting effective references from available candidates remains essential under computational constraints.

Under limited computational resources, reference image selection should primarily consider image quality, phenological similarity, and temporal proximity. Image quality is the first priority, and observations affected by severe cloud contamination or information loss should be excluded. Phenological similarity determines the consistency of surface spectral and structural characteristics; thus, in regions dominated by phenological variations, images acquired at similar phenological stages to the target phase should be preferentially selected. In contrast, for areas dominated by nonphenological changes, such as urbanization-driven surface dynamics, reference images with smaller temporal gaps from the target phase are more desirable. Such a preliminary reference image selection strategy can improve the balance between computational efficiency and fusion accuracy.

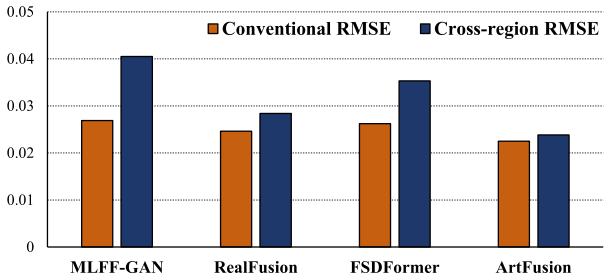


Fig. 15. Performance comparison of transferability test in the Southern Taihang Mountains area.

F. Inter-Regional Transferability Test

Inter-regional transferability remains a longstanding challenge in STF. To evaluate the robustness of ArtFusion under cross-regional deployment, the model trained on the Poyang Lake Wetland was directly applied to the Southern Taihang Mountains Area without any fine-tuning. Owing to substantial differences in atmospheric conditions, land-cover characteristics, and land-surface dynamics between the two regions, some performance degradation is inevitable for deep learning-based methods. As illustrated in Fig. 15, the degradation

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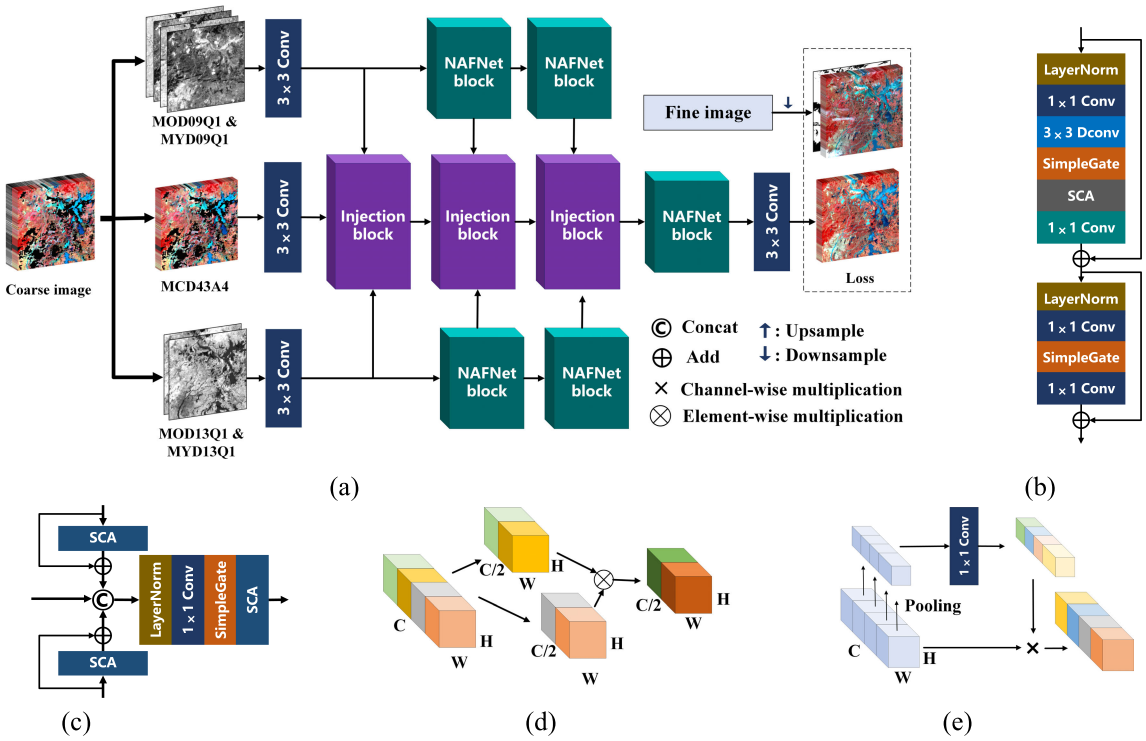


Fig. 16. Architecture of SIQRN-lite. (a) SIQRN-lite network. (b) NAFNet. (c) Injection block. (d) SG. (e) SCA.

was quantified using the relative increase in RMSE from the conventional experiments to the cross-region experiments. MLFF-GAN, RealFusion, and FSDFormer exhibited RMSE increases of 50.6%, 15.4%, and 34.7%, respectively, whereas ArtFusion showed only a 5.8% increase.

This superior transferability is consistent with the design philosophy of ArtFusion. By enabling the flexible utilization of an arbitrary number of auxiliary images through adaptive attention weighting, the proposed framework learns more generalizable spatiotemporal representations rather than relying on a fixed set of reference pairs. Together with its extremely compact architecture, ArtFusion is less susceptible to overfitting to region-specific characteristics and therefore exhibits stronger generalization capability across geographically distinct regions. These results demonstrate the considerable potential of ArtFusion for large-scale cross-regional STF applications.

G. Further Improvement of ArtFusion

Despite its superior accuracy and lightweight architecture, ArtFusion has several limitations. First, the abundance of available historical imagery poses a challenge: memory overhead increases linearly with the number of inputs, and as noted, excessive inputs can lead to spectral reconstruction performance saturation and spatial blurring. Future work should investigate lightweight prescreening mechanisms to select the most informative reference pairs before fusion, while exploring data compression techniques to further reduce memory consumption [54]. Second, MODIS restoration currently relies on a pretrained SIQRN-lite model. Developing an end-to-end framework that jointly trains the MODIS reconstruction and STF modules would enhance deployment flexibility and model transferability. In addition, although ArtFusion was validated using the Landsat–MODIS data combination, its applicability to other satellite systems, such as the GF series, requires further investigation [55]. Furthermore, reconstructing previously unseen land-surface changes remains challenging. Although ArtFusion demonstrates strong generalization capability under extreme scenarios such as flood-induced surface changes, events without historical analogs still introduce considerable uncertainty due to the lack of corresponding temporal references. In such cases, relying solely on optical observations may be insufficient. Integrating complementary multisource observations, such as SAR imagery, UAV data, or ground-based measurements, has the potential to provide additional structural and physical information for more reliable reconstruction. Accordingly, future work will focus on developing multimodal fusion frameworks capable of processing arbitrary numbers of cross-source reference inputs.

VI. Conclusion

Limited exploration of multitemporal reference-based STF has constrained the performance of existing methods, creating a research mismatch between the current technical paradigm and the wealth of available satellite archives. This study proposes ArtFusion, an arbitrary-reference STF model that achieves superior reconstruction accuracy through four architectural innovations: ECHB for deep, efficient feature extraction, ETIE to integrate acquisition temporal information, MCAF module for fine-grained multitemporal feature aggregation, and QGSG to enhance utilization of reliable spectral–spatial information.

Experimental results across two study areas demonstrate that ArtFusion consistently outperforms several SOTA

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methods (MLFF-GAN, C-ROBOT, and RealFusion) in both qualitative and quantitative comparisons, particularly in four challenging scenarios, including areas with rapid phenological changes, drastic morphological variations, highly heterogeneous landscapes, and frequent cloud contamination. By breaking the reliance on limited reference pairs and explicitly incorporating temporal information, ArtFusion unlocks the latent potential of historical satellite archives to achieve unprecedented reconstruction accuracy and reliability, establishing a new performance benchmark for STF in complex, dynamic environments.

APPENDIX

SIQRN-lite is a lightweight variant of SIQRN, obtained by removing the initial encoder–decoder stage to improve computational efficiency. Detailed descriptions of the original SIQRN are available in Guo et al. [33]. By integrating complementary information from MCD43A4, MOD09Q1, MYD09Q1, MOD13Q1, MYD13Q1, Landsat 8 OLI surface reflectance, and monthly median NDVI composites, SIQRN-lite restores coarse imagery through quality correction, spectral harmonization with Landsat, and preliminary spatial resolution enhancement, thereby reducing uncertainty in the subsequent STF process.

As illustrated in Fig. 16, SIQRN-lite employs information injection blocks to fuse multisource features and adopts the nonlinear activation free network (NAFNet) [56] as the backbone. Each NAFNet block consists of Layer Normalization, convolutional layers, the SimpleGate (SG) activation, and the simplified channel attention (SCA) module. Features extracted from MOD09Q1, MYD09Q1, MOD13Q1, and MYD13Q1 are injected into MCD43A4 feature maps through information injection blocks, and the fused features are compressed by a final 3×3 convolution to generate the restored coarse image. The network is optimized using the RMSE between the restored coarse image and the $8\times$ downsampled fine image, with cloud-contaminated pixels excluded from loss computation.

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